

Website Traffic Analysis

1. Objective

The objective of this project is to analyze website traffic data to understand user behavior, evaluate traffic sources, measure user engagement, and identify opportunities to improve website conversions using Python and Tableau.

2. Dataset Information

Dataset Name: Website Traffic Dataset

Source: Kaggle

Dataset Description

- The dataset contains information about website visitor behavior and engagement.
- It includes page views, session duration, bounce rate, traffic source, time spent on pages, previous visits, and conversion rate.
- The dataset is used to analyze website performance, user engagement, and marketing effectiveness.

Key Columns Used

- Page Views
- Session Duration
- Bounce Rate
- Traffic Source
- Time on Page
- Previous Visits
- Conversion Rate

Total Records: 2000

Total Columns: 7

3. Data Cleaning

```
import pandas as pd

df = pd.read_csv("traffic.csv")
df.head()
```

[1]

	Page Views	Session Duration	Bounce Rate	Traffic Source	Time on Page	Previous Visits	Conversion Rate
0	5	11.051381	0.230652	Organic	3.890460	3	1.0
1	4	3.429316	0.391001	Social	8.478174	0	1.0
2	4	1.621052	0.397986	Organic	9.636170	2	1.0
3	5	3.629279	0.180458	Organic	2.071925	3	1.0
4	5	4.235843	0.291541	Paid	1.960654	5	1.0

Figure 1:Dataset Preview

```
df.info()
```

[3]

```
<class 'pandas.DataFrame'>
RangeIndex: 2000 entries, 0 to 1999
Data columns (total 7 columns):
#   Column                Non-Null Count  Dtype  
---  -
0   Page Views            2000 non-null  int64  
1   Session Duration      2000 non-null  float64
2   Bounce Rate           2000 non-null  float64
3   Traffic Source        2000 non-null  str     
4   Time on Page          2000 non-null  float64
5   Previous Visits       2000 non-null  int64  
6   Conversion Rate       2000 non-null  float64
dtypes: float64(4), int64(2), str(1)
memory usage: 109.5 KB
```

Figure 2:Dataset Information

```
df.isnull().sum()
```

[5]

```
Page Views      0
Session Duration 0
Bounce Rate     0
Traffic Source  0
Time on Page    0
Previous Visits 0
Conversion Rate 0
dtype: int64
```

Figure 3:Missing Values

4. Summary Statistics

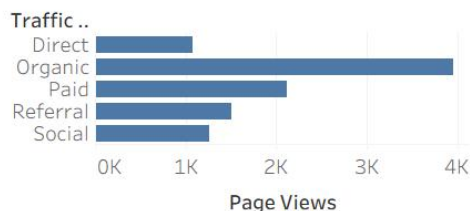

```
[15] df["Traffic Source"].value_counts()

... Traffic Source
Organic      786
Paid         428
Referral     301
Social       269
Direct       216
Name: count, dtype: int64
```

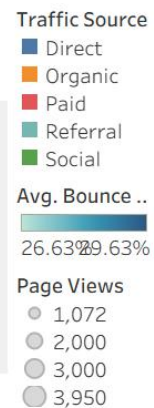
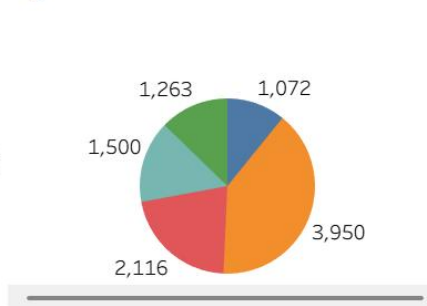
6. Visualizations

Website Traffic Analysis Dashboard

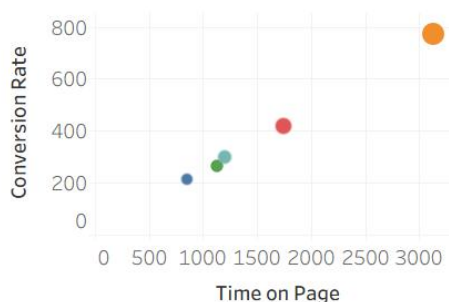
Total Page Views by Traffic Source



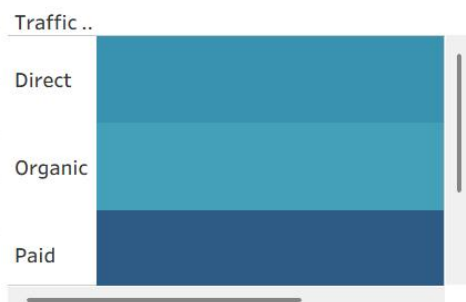
Average Session Duration by Traffic Source



Time on Page vs Conversion Rate



Average Bounce Rate by Traffic Source



Dashboard Includes:

- Total Page Views by Traffic Source (Bar Chart)
- Average Session Duration by Traffic Source (Pie Chart)
- Time on Page vs Conversion Rate (Scatter Plot)
- Average Bounce Rate by Traffic Source (Heat Map)

Observation:

- Traffic sources contribute differently to website visits.
- User engagement varies across traffic sources.
- Bounce rates differ by source, indicating varying traffic quality.
- The relationship between time on page and conversion rate helps assess user engagement and conversion performance.

7. Executive Summary

The website traffic analysis provides valuable insights into visitor behavior and website performance. By examining page views, session duration, bounce rates, and conversion rates, it becomes easier to identify which traffic sources attract engaged users and which areas require improvement. The Tableau dashboard presents these insights visually, enabling data-driven decisions to improve user experience and website performance.

8. Five Tactical Improvements

1. Focus marketing efforts on the traffic sources that generate the highest page views and conversions.
2. Reduce bounce rates by improving landing page content and website loading speed.
3. Enhance page content to encourage users to spend more time on the website.
4. Optimize high-traffic pages with stronger calls-to-action to improve conversion rates.
5. Continuously monitor website analytics to identify trends and make informed business decisions.